

# Verification is a cost/accuracy problem: *price the verdict, not the detector*

Jiawei Li · Inference Verification Gym (ivgym) · all numbers measured on one NVIDIA H100 with Qwen3-1.7B audited by a Qwen3-0.6B proxy

A client renting inference cannot see which model ran. Every defence against that substitution is reported as a detection accuracy and priced by a rate — FLOPs per sequence, seconds per prefill — and neither is the quantity a client buys. It buys a VERDICT at a false-accusation budget it can live with, and the price of a verdict factors exactly into how many tokens of evidence it needs and what a token of that evidence costs. We measure both factors for 5 deviations  $\times$  7 detectors on one model in one run, test the model that converts them into a price at 280 (cell, batch) points, and sweep the price of a token over audit batch. Measured this way the ranking of the defences changes, 14 of 35 cells have no finite price at all, and the engineering that matters moves off the estimator and onto the accounting.

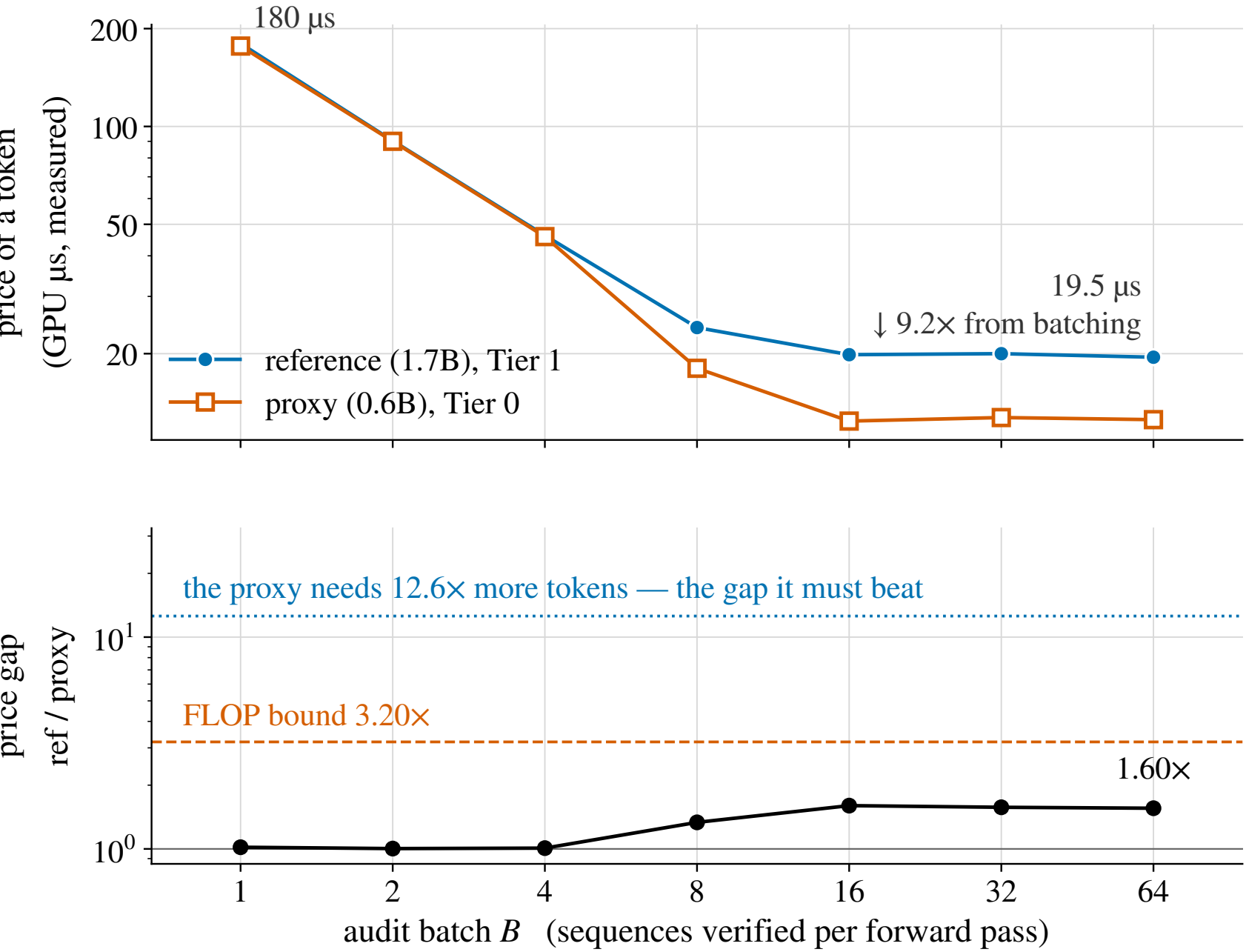
## 1 WHAT A VERDICT COSTS

$$C(\text{verdict}) = \underbrace{b^*(d')}_{\text{tokens of evidence}} \times \underbrace{c(\text{tier})}_{\text{price per token}} = (\delta^*/d')^2 \times c(\text{tier}) \quad (1)$$

$d'$  is the per-token effect size of a deviation under a detector, measured here with a sequence-level bootstrap.  $\delta^* = 3.767$  is the batch separation that reaches standardized pAUC 0.90 at FPR  $\leq 0.5\%$  — the repo's standing protocol, not a new one. Both factors are measured on the same pool ( $80 \times 256 = 20,480$  tokens per configuration), so their product is a real number and not a composition of numbers from different runs.

## 2 THE PRICE OF A TOKEN IS NOT A CONSTANT

Batch 1 is the regime a per-sequence audit runs in, and it is also the regime in which a 3.2 $\times$ -fewer-FLOPs proxy is only 1.02 $\times$  cheaper per token: a 256-token prefill reads every weight to do a few hundred token-rows of work, so the price tracks parameter bytes, not arithmetic. Swept over audit batch, the price of a token falls 9.2 $\times$  — an order of magnitude more than the choice of tier ever buys — while the tier gap saturates at 1.60 $\times$ , half of its own FLOP bound and far below the 12.6 $\times$  token penalty the proxy pays for the same verdict.



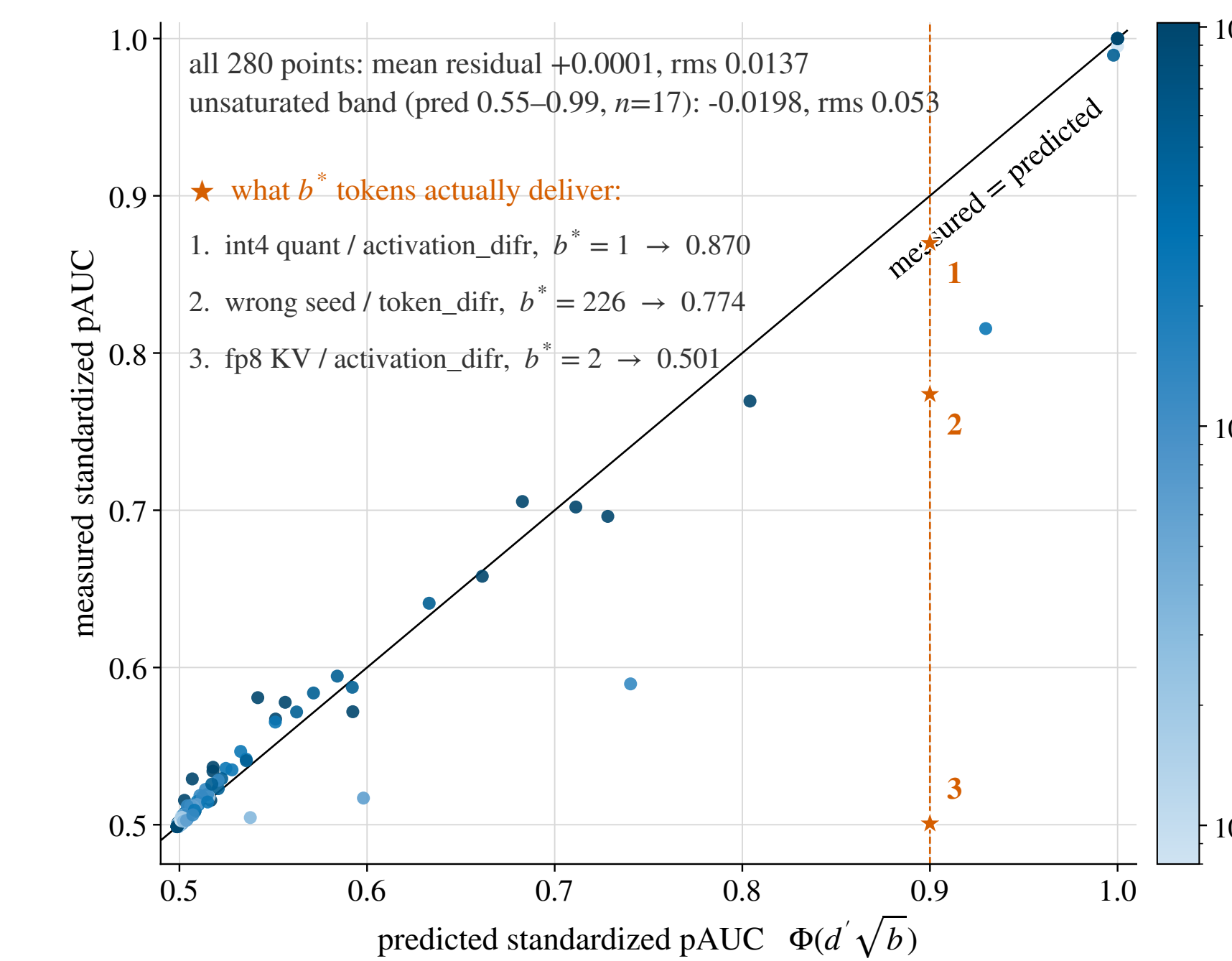
**Fig. 11** Measured GPU-synchronised price of one token of evidence against audit batch (top), and the Tier-1/Tier-0 price gap against the two bounds that decide whether the cheap tier is worth buying (bottom). 9 timed repeats after 3 warmups, bfloat16, eager attention, full logits kept, 256 tokens per row. Batching is the only lever in this figure that moves cost by an order of magnitude. Dotted line: the token penalty from Table 1.

## 3 PROTOCOL

One protocol produces every number. Detection is the standardized partial AUC at FPR  $\leq 0.5\%$ ; the threshold comes from a held-out honest calibration split; per-token scores are winsorized at their 99.9th honest percentile; the batch/pool ratio is capped at 10% and the cap is enforced in code, because batches resampled from a finite pool inflate an AUC without leaving a sign.  $d'$  carries a 400-resample sequence-level bootstrap interval — tokens inside a sequence are not independent, and a token bootstrap would understate the error on a price that goes as  $1/d'^2$ . Prices are GPU-synchronised wall-clock on an idle H100, each stage timed alone with the reference cache dropped, converted at \$2.50/hour. A cell whose  $d'$  does not clear its OWN detector's honest-vs-honest control is void, not cheap.

## 4 IS THE PRICE ITSELF TRUSTWORTHY?

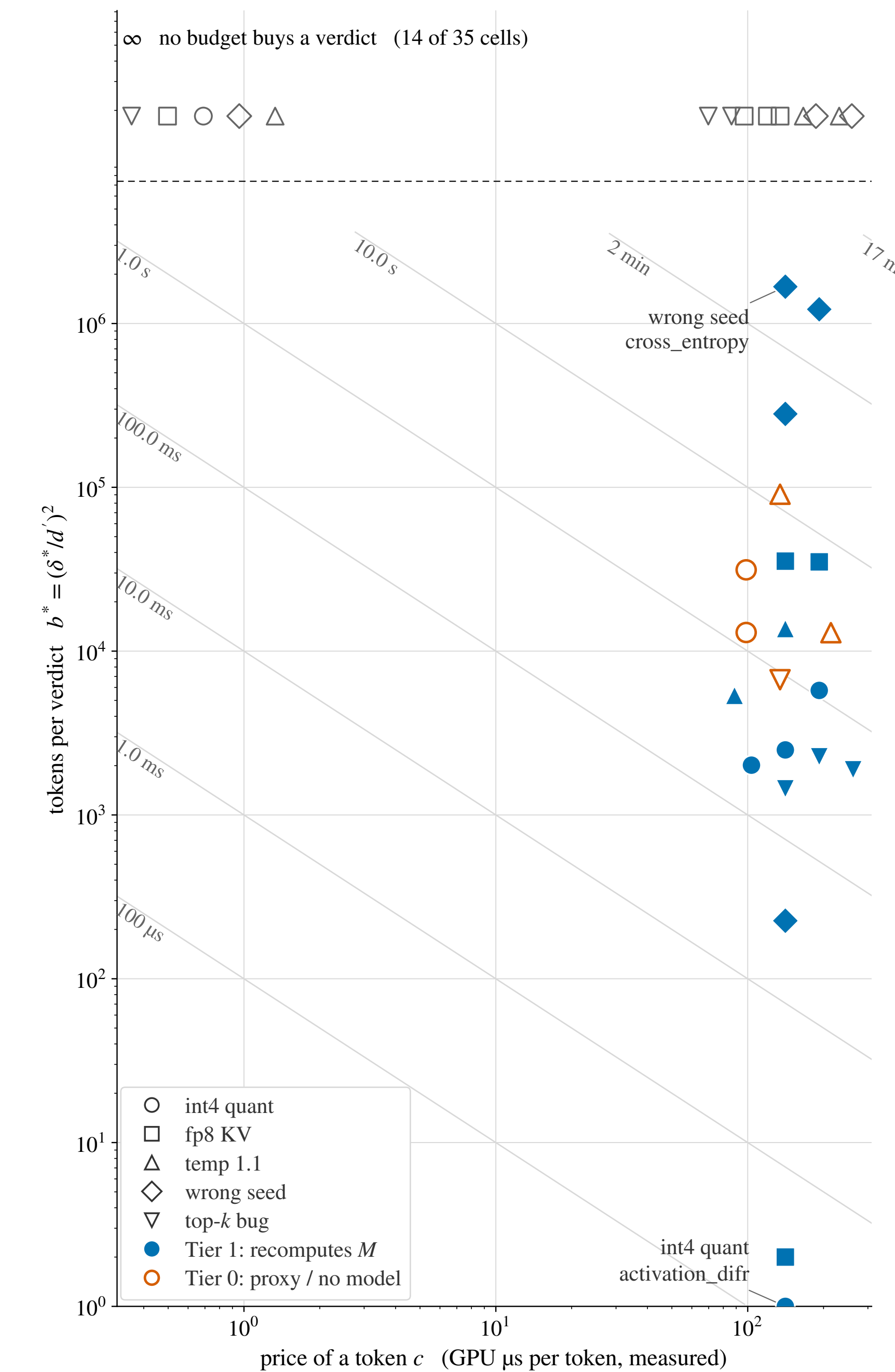
Half of eq. (1) is a prediction: the CLT says  $b$  independent tokens separate by  $d'\sqrt{b}$ . Two documented effects break it — tokens inside a sequence are correlated, and a heavy-tailed per-token score reaches the Gaussian limit slowly — so we test it on the same score arrays the prices come from. Over all 280 points the model is unbiased (mean residual +0.0001), but that average is carried by saturated cells; in the unsaturated band it is OPTIMISTIC by 0.020 pAUC, and at the one cell where the 0.90 crossing is directly observable the verdict costs 1.59 $\times$  the tokens it was priced for. Every price in §6 is a LOWER BOUND on cost.



**Fig. 21** Predicted versus measured standardized pAUC, 35 cells  $\times$  8 batch sizes, every point re-measured under the protocol of §3 at 4,000 batches per point. Stars: the pAUC that  $b^*$  tokens actually deliver, at the 3 cells whose  $b^*$  fits under the 10% pool ceiling (max legal batch 1,024); the cheapest cell on the board delivers 0.870, not 0.90. Every other price in Table 1 is an extrapolation from this checkable range and is labelled one.

## 5 COST AND ACCURACY DO NOT TRADE OFF ALONG A CURVE

Cost/accuracy is usually drawn as a Pareto curve of rates: cheap detector, weak; expensive detector, strong. It is not a curve of rates. Each mark below is one deviation scored by one detector, the axes are the two measured factors of eq. (1), and the product — the cost of a verdict — is carried by the diagonals. A detector 12 $\times$  cheaper per token that needs 400 $\times$  more tokens is 30 $\times$  more expensive per verdict, and a detector with  $d' \leq 0$  is not on the curve at all.



**Fig. 31** The cost plane, log-log. Grey diagonals are iso-cost: every point on one buys a verdict for the same GPU time. Marker shape is the deviation, colour and fill the tier. Measured price takes only three values — a tier runs a model or it does not — and the two that run one price within 1.05 $\times$  of each other at batch 1, so the vertical axis decides almost everything; marks are nudged sideways only where they would overlap. Above the dashed break, the rail of cells no budget reaches: 13 at  $d' \leq 0$  and 1 whose  $d'$  does not clear its own detector's honest-vs-honest control.

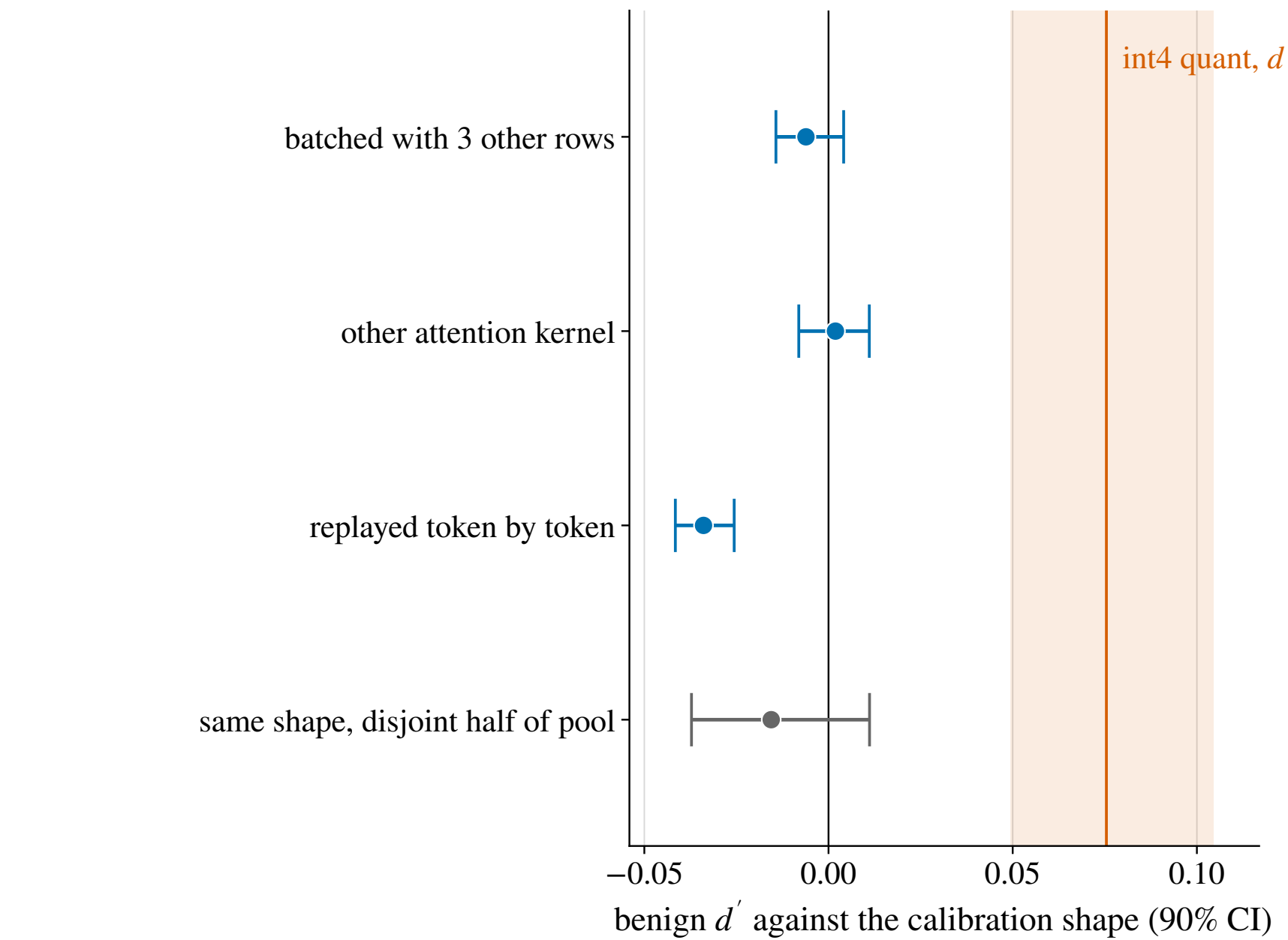
## 6 THE GRID, PRICED

|                 | Tier 1 — recomputes $M$ |               |               |                              | Tier 0 — never runs $M$ |               |                  |
|-----------------|-------------------------|---------------|---------------|------------------------------|-------------------------|---------------|------------------|
|                 | token DiFR              | cross entropy | top-loc       | activation DiFR              | accept rate             | surface stat  | surface tokens   |
| price per token | 141.1 $\mu$ s           | 141.1 $\mu$ s | 141.1 $\mu$ s | 141.1 $\mu$ s                | 134.4 $\mu$ s           | 134.4 $\mu$ s | 0.7 $\mu$ s      |
| int4 quant      | <b>352 ms</b>           | <b>812 ms</b> | <b>284 ms</b> | <b>141 <math>\mu</math>s</b> | 4.2 s                   | 1.7 s         | $\infty$         |
| fp8 KV          | 4.9 s                   | $\infty$      | 5.0 s         | 282 $\mu$ s                  | $\infty$                | $\infty$      | $\infty$         |
| temp 1.1        | $\infty$                | 758 ms        | 1.9 s         | $\infty$                     | 12.2 s                  | 1.7 s         | $\infty$         |
| wrong seed      | 31.9 ms                 | 4 min         | 39.6 s        | 3 min                        | $\infty$                | $\infty$      | $\infty$         |
| top-k bug       | 266 ms                  | 319 ms        | 204 ms        | $\infty$                     | $\infty$                | 899 ms        | $\infty^\dagger$ |

**Table 11** GPU time for one verdict, 5 deviations  $\times$  7 detectors, one model, one run.  $\infty$  = no batch size reaches pAUC 0.90;  $\dagger$  = a finite  $b^*$  that does not clear the detector's own honest control, which is not a price. Bold: the cheapest verdict on the board, 141  $\mu$ s on 1 token(s) of activation DiFR — which pays for it in ACCESS, not compute: it needs the provider's internal activations, which a rented endpoint does not return. The marginal verdict is priced here; the honest calibration pool it is scored against ( $20 \times b^*$  tokens) amortizes over verdicts and is reported separately.

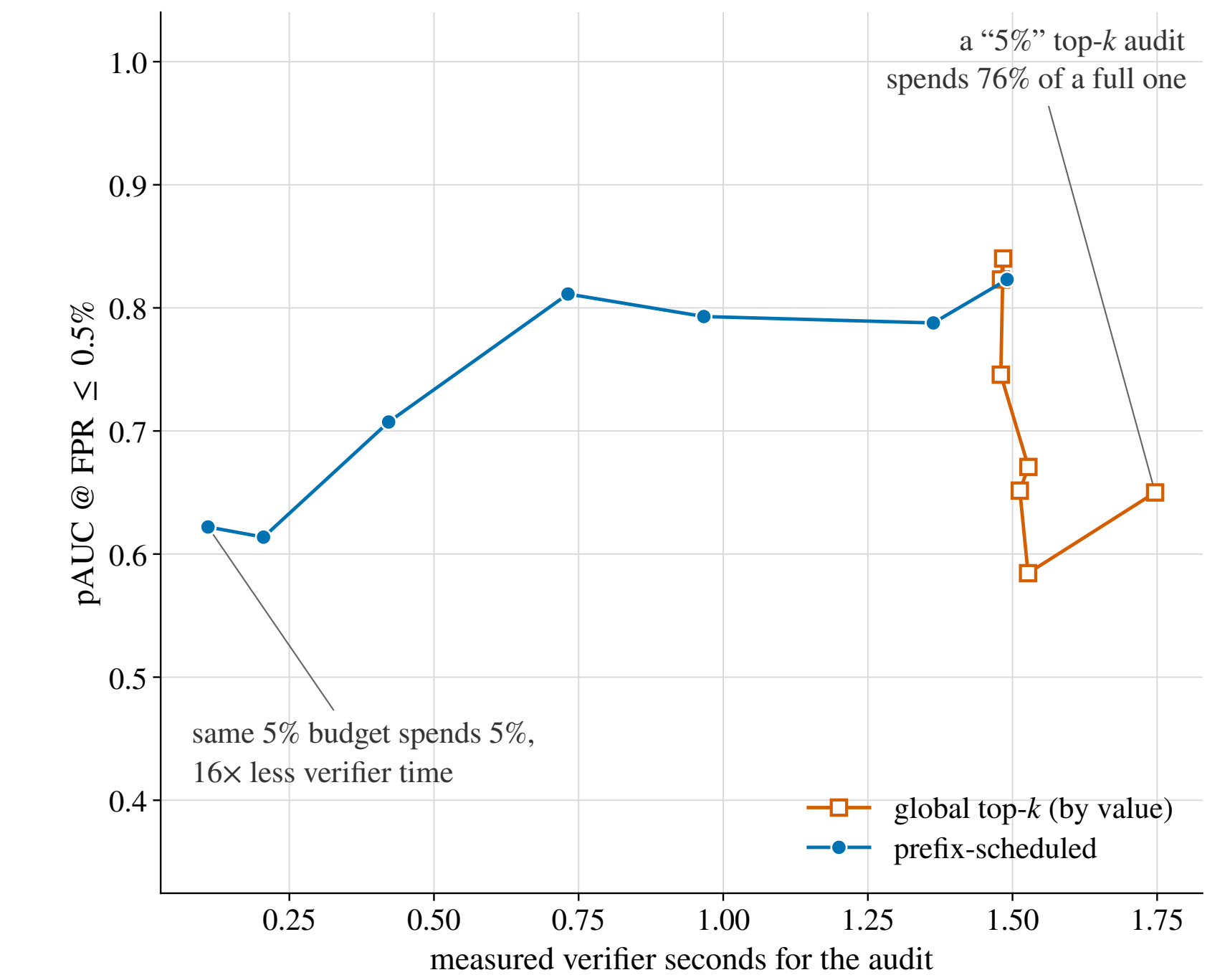
The cheap tier is not a weak detector — it is a detector of a different thing. On model SUBSTITUTION the same proxy reads AUC 0.998 at 2.3 $\times$  fewer FLOPs than recomputation while never running  $M$ . On the numerical deviations in Table 1 it needs 12.6 $\times$  more tokens than a recompute for the cell it can price at all, and has no finite price for three of the five.

## 7 THE ACCURACY NO BUDGET BUYS



**Fig. 41** Same weights, same tokens, same noise draw — only the verifier's own forward-pass schedule differs, and every row is honest. Replaying the SAME shape returns  $d' = 0.000000$  exactly. A benign floor reaching the orange rule (int4 quant,  $d' = 0.0754$ , band = its 90% CI) would push that verdict's price to infinity. None does — the largest is +0.0019 — but token-by-token replay moves the null 45% as far as the attack does, away from it. Calibrate in the shape you audit in.

## 8 CHEAPER TOKENS: SCHEDULE AGAINST THE REAL COST



**Fig. 51** Reading  $M$  at generated position  $j$  requires a prefill of everything before it, so the physical budget is PREFILL tokens: depth is cheap, breadth is expensive. Scheduling the same audit budget prefix-wise buys the same accuracy for 16 $\times$  less verifier time. No statistics in it — it replicates across deviations by construction.

## 9 WHAT WE WOULD DEPLOY

- Price the verdict, not the detector.**  
14 of 35 cells cost infinity; the cheapest finite one, 141  $\mu$ s, is only available to a verifier with activation access.
- Buy the exponent, not the constant.**  
Tokens go as  $1/d'^2$  and price per token as model size, so effect size wins every time the two compete.
- Batch the audit before switching tiers.**  
Batching buys 9.2 $\times$ ; the cheap tier buys 1.60 $\times$  and costs 12.6 $\times$  the tokens.
- Treat a predicted price as a lower bound.**  
Where it can be checked, the model is optimistic by 1.59 $\times$  in tokens.
- Measure your own benign floor first.**  
It is the price ceiling: no budget buys accuracy below it, and it is a property of your serving stack, not of the attack.
- Stop early, and size the pool to say so.**  
Sequential testing saves 1.3 $\times$  at matched power, but only 3 of 22 cells have a pool deep enough to demonstrate it.

## 10 WHAT THIS DOES NOT SHOW

### The deviations are logit-level models.

They are logit-space stand-ins for quantisation and sampling faults, not quantised checkpoints re-served through a real stack.

### One prompt domain, one model pair, one accelerator.

The cross-GPU half of the benign floor is unmeasured, so the rows of Fig. 4 are a lower bound on it.

### Two transformers builds, so compare ratios and not levels.

Fig. 1 re-times the tiers under a newer build than the batch-1 prices of Table 1 and runs 1.28 $\times$  slower in absolute terms; only within-figure ratios carry across the two.

### Table 1 prices the marginal verdict.

It excludes the honest calibration pool, which amortizes over verdicts, and 22 of 35 cells cannot be checked at  $b^*$  under the 10% pool ceiling.